Large Data Sets

Questions 1–80

1.58 — Question 1. State the following facts about the Large Data Set:

a) The 5 locations in the UK.
b) The 3 locations not in the UK.
c) The years in which the data was collected.
d) The months which are included and the maximum total number of days in those months.

1.58 — Question 2. Which of the locations

a) are coastal?
b) are south of the equator?

1.58 — Question 3. Why does Perth have a lower temperature than Jacksonville for the data in the Large Data Set?

1.58 — Question 4. Which of the UK locations has the highest windspeed?

1.58 — Question 5. State the units of the following:

a) Daily mean temperature.
b) Daily total rainfall.
c) Daily total sunshine.
d) Daily maximum relative humidity.
e) Daily mean windspeed.
f) Cloud cover.
g) Visibility.
h) Pressure.

1.58 — Question 6. Convert the following:

a) 3 knots into mph.
b) 5.75 mph into knots.
c) 30 mph into knots.

1.58 — Question 7. Explain what it means for cloud coverage to be

a) 1 oktas

b) 4 oktas

c) 8 oktas

1.58 — Question 8. According to the Large Data Set, on 29th May 1987, the wind in Heathrow was blowing with bearing 290°. In which direction was the wind blowing as a compass direction?

1.58 — Question 9. In the Large Data Set, is a location more likely to be high cloud coverage or low cloud coverage?

1.58 — Question 10. Zara has calculated the mean and standard deviation of a sample of rainfall in Leeming in September 2015.
She used only the data which has a value, obtaining $\bar{x} = 1.6$ and $\sigma = 4.2$ where $n = 24$

a) Comment on the validity of her results.
b) Comment on how the mean and standard deviation will change after a correction.

1.58 — Question 11. Two months' worth of data for hours of daily sunshine, x has been summarised for Hurn for July & August 2015 below:

$$\sum x = 321.2, \quad \sum x^2 = 2692.66, \quad n = 62$$

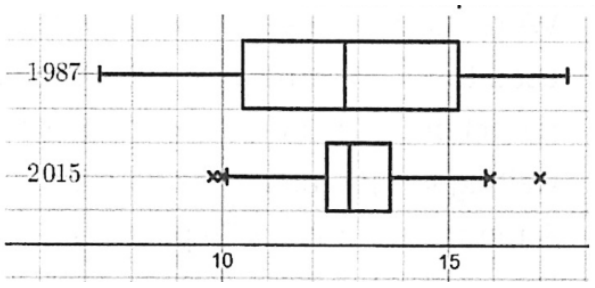
a) Calculate the mean and the standard deviation, where $s_x = \sqrt{\dfrac{\sum x^2 - \frac{(\sum x)^2}{n}}{n-1}}$, giving your answer to 2 decimal places.

In Hurn for the same months in 1987 gives

$$\bar{x} = 6.42 \text{ and } s_x = 4.62$$

b) Compare these to the statistics calculated for 2015 above.

1.58 — Question 12. The daily temperature for the same two consecutive months in both 1987 and 2015 in Camborne are shown in the boxplots below:



a) Which are these months likely to be?

It is claimed based on these data that the mean temperature is becoming more consistent across the UK.

b) Explain the validity of this claim.

1.58 — Question 13. Jeremy is investigating the daily total rainfall in Jacksonville in October 2015. He selects data by selecting the first data point, then picks every 6th one after that until he reaches the end of the data set. State

a) State the type of sampling Jeremy used.

b) State a mistake with his application of this sampling method.

1.58 — Question 14. The table below shows the probability distribution for c, the cloud coverage in Leeming in 2015.

c	1	2	3	4
$\mathrm{P}(X=c)$		0.04	0.09	0.14
c	5	6	7	8
$\mathrm{P}(X=c)$	0.17	0.22	0.23	

a) Find the missing probabilities given it is known that

$$10\mathrm{P}(c=1)=\mathrm{P}(c=8)$$

b) Find the expected number of days that the cloud coverage is 3 oktas or less.

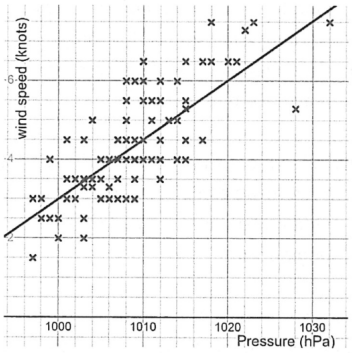
It is claimed that this data could be modelled by $X \sim \mathrm{B}\left(54, \frac{1}{9}\right)$

c) Using this model, find the number of days that cloud coverage is 3 oktas or less.

d) Comment on the validity of this claim.

1.58 — Question 15. A sample of 100 data points has been taken from Beijing in 1987. The windspeed w (knots) has been plotted against pressure p (hPa) and is shown in the diagram below. A regression line has been found, given by

$$w = 0.151p - 148$$

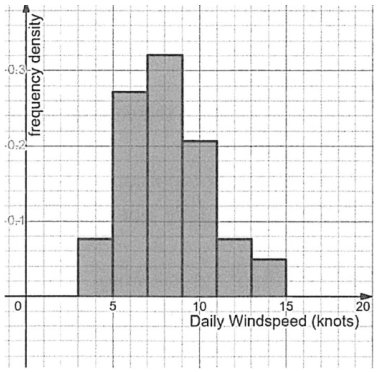


a) Use the model to predict the windspeed when there is a pressure of 1026 hPa.

The PMCC is found to be $r = 0.734$

b) Comment on why there is reason to believe that the data may not have been sampled randomly.

1.58 — Question 16. Prerequisite knowledge for Q16: Normal Distribution The daily windspeed in Perth in 2015 is shown on the histogram below:



The heights of the bars are $\{0.076, 0.272, 0.321, 0.206, 0.076, 0.049\}$

a) Estimate the mean and the standard deviation of the data.

b) Estimate the number of days the windspeed is above 12 knots.

c) By modelling the data using a normal distribution with your mean and standard deviation, find the probability of the windspeed being over 12 knots.

d) Is the normal distribution a suitable model for these data?

1.58 — Question 17. Prerequisite knowledge for Q17: PMCC Hypothesis Testing Adrian believes that there is a negative correlation between the windspeed and relative humidity in Leeming in 2015.
He collects a random sample of 20 dates and calculates the sample's correlation coefficient to be -0.3613
The critical value for $n = 20$ for a one-tailed test at a 5% significance level is 0.3783.

a) Carry out a hypothesis test at the 5% significance level to investigate whether Adrian's belief is supported.
b) A second sample of 20 dates is taken, and the correlation coefficient is -0.6234. Write the conclusion of the hypothesis test for this sample.

1.59 — Question 1. State the following facts about the Large Data Set:

a) The five makes of vehicle.
b) The years in which the vehicles were registered.
c) The regions in which the registered keepers live.
d) The years in which the vehicles were manufactured.
e) The 5 types of propulsion.
f) The 9 types of body type.

1.59 — Question 2. State the units of the following:

a) CO_2 (carbon dioxide).
b) CO (carbon monoxide).
c) NOX (oxides of nitrogen).
d) part (particulate emissions).
e) hc (hydrocarbon emissions).
f) mass.
g) engine size.

1.59 — Question 3. Explain why, when sampling from the Large Data Set, that blanks may be found in some the data.

1.59 — Question 4. Kerry samples a car from the Large Data Set and notes down that its mass, particulate emissions, and engine size are all 0.
In each case, state whether Kerry may have made a mistake or has correctly noted the details of the car.

1.59 — Question 5. For each of the following, given a sample is being taken from the Large Data Set, write down the name of the sampling method used.

a) The order of the cars is randomised, then every 100th is noted.
b) The order of the cars is randomised, then the first 10 of each type of make is noted to obtain 50 cars.
c) The data set is sorted by makes and the proportions of each type calculated. Using a random number generator and the reference number, each make is selected to the same proportion as the full data set.

1.59 — Question 6. Martha uses the complete Large Data Set and finds that the mean mass of a car in 2002 is 1278.7 kg and in 2016 is 1416.6 kg. She finds out that the average mass of a human in 2002 and 2016 was 75.9 kg and 78.0 kg respectively.

a) She says the actual mass of the cars is 1202.8 kg and 1338.6 kg. Explain why Martha is incorrect.

b) She claims that all vehicles in the UK are heavier in 2016 than in 2002. Comment on the validity of Martha's claim.

1.59 — Question 7. The table below shows a sample of 10 vehicles from the Large Data Set.

Make	Region	Engine Size
Vauxhall	London	1398
Volkswagen	London	1198
Ford	North West	999
Volkswagen	London	1598
Vauxhall	South West	1398
Ford	London	998
Toyota	London	1398
Vauxhall	South West	1398
BMW	London	2171
Ford	North West	1498

a) Find the mean and the standard deviation of the engine size of this sample.

b) Give a reason why the sample used above may have biased data.

Any value more than 2 standard deviations from the mean can be identified as an outlier.

c) Find any outliers in this sample of engine size.

The mean engine size for the whole of the Large Data Set is 1646.2 cm^3 and the standard deviation is 456.0 cm^3

d) Give one comparison of central tendency for this sample vs the full data set.

e) Give one comparison of spread of this sample vs the full data set.

1.59 — Question 8. The table below shows the CO_2 emissions (g/km) in 2002 and 2016, for all the diesel cars in the Large Data Set.

	2002	2016
$\sum x$	46 335	127 096
Number of cars	302	1 097

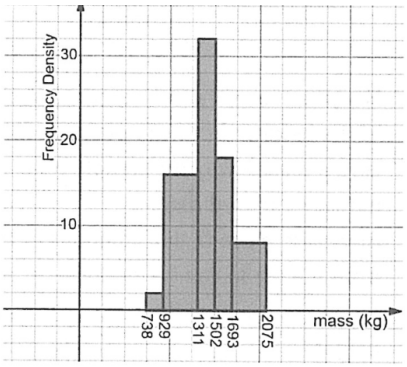
a) Find the mean of the CO_2 emissions for 2002 and for 2016

The mean CO_2 emissions for all the cars registered in 2016 in the Large Data Set is 120.37. There are 2529 total cars in the Large Data Set in the year 2016.

b) Find the mean of all the CO_2 emissions for all the vehicles in 2016 which are not diesel.

c) It is claimed that the increase in CO_2 emissions is caused by the electric cars which feature solely in the 2016 data. Explain, with reference to the Large Data Set, why this cannot be the case.

1.59 — Question 9. A set of cars were randomly sampled from the Large Data Set and the masses have been displayed in the histogram below.



It is known that the bar between 1693 kg and 2075 kg represents 8 cars.

a) Calculate how many cars have been sampled altogether.

b) Estimate the mean mass of a car in this sample.

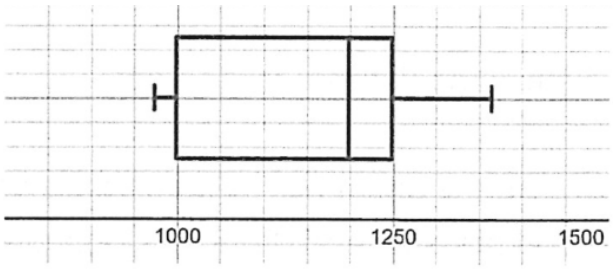
c) Find the probability of picking a car of mass between 1120 kg and 1693 kg.

1.59 — Question 10. The Keeper Title of a sample of 250 cars are taken from the Large Data Set and a probability distribution is given:

Keeper Title	Probability
Male	$\frac{2}{5}$
Female	k
Unknown	$\frac{4}{125}$
Company	$\frac{9}{25}$

a) Find the value of k
b) State the number of each of the Keeper Title sampled.
c) Give one reason why the sample is representative of the Large Data Set.

1.59 — Question 11. A set of 100 cars were sampled from the Large Data Set and the engine sizes of these cars are displayed in the boxplot below:



Using your knowledge of the Large Data Set, explain why it is possible that it was not sampled randomly.

1.59 — Question 12. Prerequisite knowledge for Q12: Normal Distribution The CO_2 emissions, x g/km, from a sample of 40 cars are taken from the Large Data Set.
The summary statistics are as follows:

$$\sum x = 5592, \qquad \sum x^2 = 802\,846$$

a) Calculate the mean and the standard deviation, where $s_x = \sqrt{\dfrac{\sum x^2 - \frac{(\sum x)^2}{n}}{n-1}}$, giving your answer to 4 significant figures.

This data is believed to be normally distributed. Find the probability that the CO_2 emissions are

b) greater than 150 g/km.

c) exactly 114 g/km.

1.60 — Question 1. These exercises were written using LDS6 but should continue to be of use for future Health Large Data Set samples. State the following facts about the Large Data Set:

a) The age of the youngest person.

b) The age of the oldest person.

c) The six categories of marital status.

d) What the category "Food30" is recording.

e) What the category "Arm" is recording.

f) The country in which the original survey is conducted.

1.60 — Question 2. State the units of the following:

a) Weight.
b) Height.
c) Body mass index.
d) Lengths of body parts.
e) Pulse rate.
f) Blood pressure.

1.60 — Question 3. What is the data set taken from?

1.60 — Question 4. Explain the difference between the systolic blood pressure and diastolic blood pressure.

1.60 — Question 5. Explain why there are columns in the Large Data Set labelled with a 1, 2, 3, 4, in the context of what was happening on the day that the data was collected.

1.60 — Question 6. What data do the columns labelled Systolic and Diastolic represent in the Large Data Set?

1.60 — Question 7. Mustafa wants a sample of 20 people from the Large Data Set.
He claims that all he needs to do is pick 20 random numbers from between 1 and 200 and use those rows from the spreadsheet.

a) State what type of sampling he has chosen
b) Explain why this might not end up with 20 useful sets of data.

1.60 — Question 8. Kristina would like to clean the data set before she samples from it. She decides to remove any rows which contain a #N/A except the 'Systolic4' and the 'Distolic4' columns. Explain the effect of this on the data set.

1.60 — Question 9. Katie would like to take a sample to do a comparison of body mass index for different categories of marital status. She would like a sample of 50 with each marital status proportionally represented.

a) State the type of sampling she should use.

b) Explain how she would do this.

1.60 — Question 10. The table below shows a sample of 10 people from the Large Data Set.

Sex	Height
Female	173.4
Female	171.4
Male	183.5
Female	174.7
Male	180.1
Male	184.9
Male	186.2
Male	173.3
Male	159.4
Male	188.4

a) Find the mean and the standard deviation of the height of this sample.
b) Give a reason why the sample used above may have biased data.
c) Any value more than 2 standard deviations from the mean can be identified as an outlier. Find any outliers in this sample of heights.

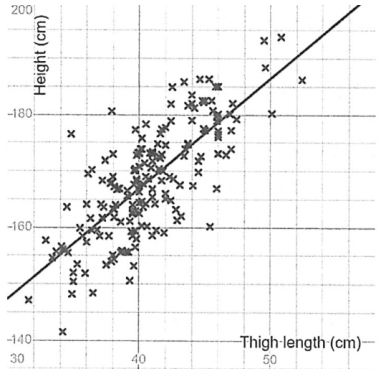
The mean height for the whole Large Data Set is 168.0 cm^3 and the standard deviation is 9.9 cm^3.

d) Give one comparison of central tendency for this sample vs the full data set.
e) Give one comparison of spread of this sample vs the full data set.

1.60 — Question 11. The scatter graph below shows the height against thigh length of the 179 data points in the Large Data Set. A regression line has been found, given by

$$h = 1.95t + 88.70$$

where h is the height of a person in cm and t is the thigh length of a person in cm.



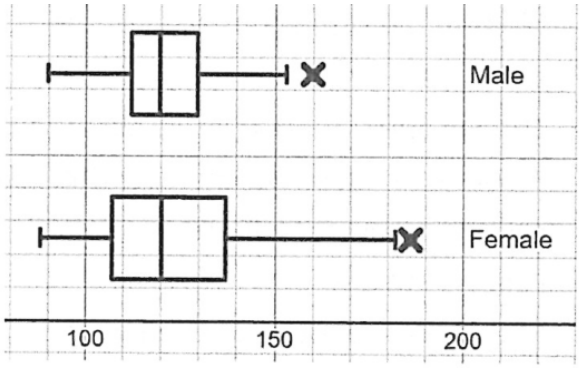
One of the people not included in the data set is 116.3 cm tall.

a) Assuming you can make a prediction by using the regression line, calculate the thigh length of this person, giving your answer to 1 decimal place.

The product moment correlation coefficient for the data is 0.7518

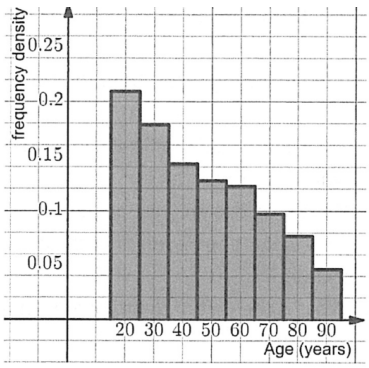
b) Explain why it is reasonable to model the relationship between t and h as a linear graph.

1.60 — Question 12. The average diastolic blood pressure for males and females is shown in the boxplots below.



a) Estimate the median of males and the median of females.
b) Estimate the interquartile range of males and the interquartile range of females.
c) Use your answers to compare the average diastolic blood pressure for males and females in the Large Data Set.

1.60 — Question 13. The histogram below shows the ages, in years, of the people in the Large Data Set.



The heights of the bars are $\{0.209, 0.179, 0.143, 0.128, 0.122, 0.097, 0.076, 0.046\}$

a) Which class interval contains the median?
b) Estimate the mean.

An adult is selected at random.

c) Find the probability that they are younger than 45 years old.
d) Explain why the final bar may not be a suitable width to represent the data.

1.60 — Question 14. Prerequisite knowledge for Q14: PMCC Hypothesis Testing Emily believes that there is a positive correlation between BMI and average systolic blood pressure. She collects a random sample of 20 people from the Large Data Set and calculates the sample's correlation coefficient to be 0.8159. The critical value for $n = 20$ for a one-tailed test at a 5% significant level is 0.3783

a) Carry out a hypothesis test at the 5% significance level to investigate whether Emily's belief is supported.

Emily's teacher is suspicious of these results, and they create a scatter graph for the whole data set, drawing a vertical and horizontal line going through the mean point. They state that "there are approximately the same number of data points in each quarter of the graph."

b) Explain why the teacher's graph shows and why they believe Emily selected the data with bias.

1.61 — Question 1. These exercises were written using LDS7 but should continue to be of use for future World Bank Large Data Set samples. State the following facts about the Large Data Set:

a) The country is the official name used by which country?
b) Which regions are included in the data?
c) How is the population of each country found?
d) What is the unit of the total area of a country?

1.61 — Question 2. Explain what a birth or death rate per 1000 is and what point of the year it is based on.

1.61 — Question 3.

a) Explain what GDP is.

b) What does 'real GDP per capita' mean?

1.61 — Question 4. Explain what is meant by 'physician density' and what definition a physician has for this data set.

1.61 — Question 5.

a) The World Health Organisation categorises healthcare workers as what?

b) According to the World Health Organisation, fewer than how many healthcare workers per 1000 would be insufficient to achieve coverage of primary healthcare needs?

Using your answer to part b)

c) Identify one of the 3 countries in Europe that do not meet the World Health Organisation value for coverage of primary healthcare needs.

d) Identify the region of the world in which no countries meet the World Health Organisation value for coverage of primary healthcare needs.

1.61 — Question 6. Explain what is meant by 'current health expenditure' and why it is a useful piece of data.

1.61 — Question 7. Caspian claims that the world has more mobile phones than people.

a) Explain how the Large Data Set indicate that this may be the case in some countries.

b) In which region are there the fewest mobile phone subscribers per 100 people in the population?

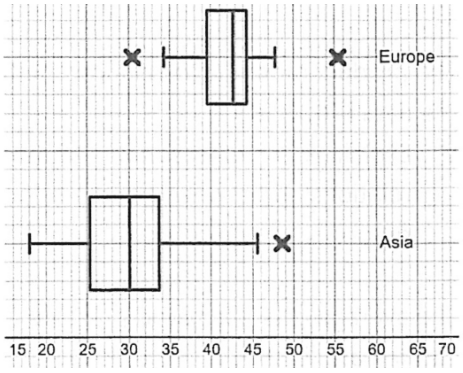
Caspian would like a sample of 46 countries to test his theory.

c) Explain how he could take a sample which proportionally represents each of the regions in the world.

1.61 — Question 8. Jamie is comparing the life expectancy in 2020 to the life expectancy in other years. Are there any countries in which life expectancy went down in the time frame from

a) 1960 to 2020?
b) 1990 to 2020?

1.61 — Question 9. The median age of countries in Europe and Asia are shown in the boxplots below:



a) Identify the outliers in each of the boxplots.
b) The Europe data had to be cleaned before it could be used, explain why this is the case and how it could be identified in the Large Data Set.
c) Make a comparison of the two data sets.

Oscar states that there is a country missing from the boxplots.

d) Which country was not included in either boxplot, but lies in the correct part of the world?

1.61 — Question 10. The table below shows the current health expenditure for the Caribbean after it has been cleaned:

Country	CHE (%)
Antigua and Barbuda	4.4
The Bahamas	5.8
Barbados	6.3
Cuba	11.3
Dominica	5.5
Dominican Republic	5.9
Grenada	5
Haiti	4.7
Jamaica	6.1
Saint Kitts and Nevis	5.4
Saint Lucia	4.3
Saint Vincent and the Grenadines	4.8
Trinidad and Tobago	7

a) Find the mean and the standard deviation of the current health expenditure for the Caribbean.

Any value more than 2 standard deviations from the mean can be identified as an outlier.

b) Identify any countries which are outliers.

The current health expenditure in Oceania is

$$\bar{x} = 8.79\% \text{ and } s_x = 5.87\%$$

c) Compare these to the statistics calculated for the Caribbean above.

d) Give two reasons why it may be fair to compare the Caribbean with Oceania.

e) Give a reason why it may not be fair to compare the Caribbean with Oceania.

1.61 — Question 11. The life expectancy at birth, L, between 1960 and 2020 can be modelled using a straight line where t is the number of decades since 1960.

$$\begin{aligned} \text{Botswana:} \quad & L = 2.01t + 50.1 \\ \text{Chile:} \quad & L = 3.90t + 55.5 \end{aligned}$$

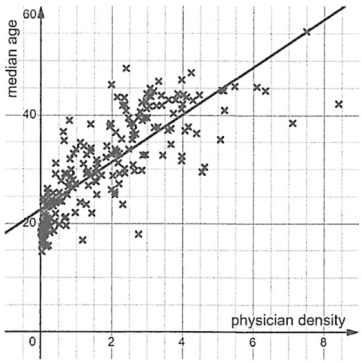
a) Predict the life expectancy for 1995 for both countries.

b) Explain what each of the values in the models mean for each country.

c) Compare the change in life expectancy of the two countries.

1.61 — Question 12. The median age of a country is plotted against the physician density (physicians/1000 population) of a country, creating the graph below. The product moment correlation coefficient is $r = 0.7949$ and the least squares regression line is given by

$$y = 4.37x + 22.6$$



a) Comment on the value of r and what it means for the association between median age and physician density.

It is known that Cuba has a very high physician density but not a high median age.

b) Identify which is likely to be the point which represents Cuba.

A country which was not included in the graph decides to manage their physician density, so it is 7 per 1000 people with the aim to increase the median age of people in their country.

c) Use the regression line to determine the median age for this country.

d) Comment on your result in part c) with reference to the aim of the country.

1.61 — Question 13. Prerequisite knowledge for Q13: Normal Distribution The death rate per 1000 people across the world may be considered normally distributed, where $X \sim \mathrm{N}(7.5, 6.92)$. Find

a) the probability that the death rate per 1000 people of a country is higher than 6

b) $\mathrm{P}(\mu - 2\sigma < X < \mu + 2\sigma)$

c) the death rate per 1000 for the 10% of the countries with the highest death rate.

1.62 — Question 1. These exercises were written using LDS8 but should continue to be of use for future London Large Data Set samples. State the following facts about the Large Data Set:

a) How many boroughs of London are listed.
b) How many areas of England are recorded (ignoring categories, Wales, England, England and Wales, and UK).
c) Who defines which boroughs are Inner or Outer London for this data set?

1.62 — Question 2. Explain what each of the following categories in the Large Data Set are:

a) Employment Rate (age 16 to 64).
b) Mean and median Income of Taxpayers (£).
c) All pupils at end of KS4 achieving 5+ A*-C including English and Maths.
d) Household Waste Recycling Rate.

1.62 — Question 3. Which properties are excluded from the median house price category?

1.62 — Question 4. Why should the data related to the mean and median income of taxpayers be treated with caution?

1.62 — Question 5. A sample of 5 London boroughs are being sampled from the Large Data Set. For each of the following, write down the name of the sampling method used.

a) Each borough is given a number between 1 and 33 then a random number generator is used to select 5, until 5 unique boroughs have been identified.

b) The order of the boroughs is randomised, then every 6th is noted.

c) The order of the boroughs is randomised, then the first 5 are sampled.

d) The data is sorted into inner and outer London and then 1 inner London and 4 outer London boroughs are randomly selected using a similar approach to in part a).

1.62 — Question 6. The table below a sample of 10 boroughs in London and their median house price in 2021:

Area	Inner or Outer London	Median House Price (£)
Barnet	Outer	575 500
Brent	Outer	525 000
Camden	Inner	765 000
Ealing	Outer	522 000
Hammersmith and Fulham	Inner	780 000
Harrow	Outer	535 000
Kensington and Chelsea	Inner	1 250 000
Merton	Outer	545 000
Richmond upon Thames	Outer	715 000
Wandsworth	Inner	656 178

a) Find the mean and the standard deviation of the median house price.

b) Give a reason why the sample used above may have biased data.

Any value more than 2 standard deviations from the mean can be identified as an outlier.

c) Identify any boroughs which are outliers.

The same boroughs in 2007 for median house price x gives

$$\bar{x} = £566\,727.50 \text{ and } s_x = £176\,435.80$$

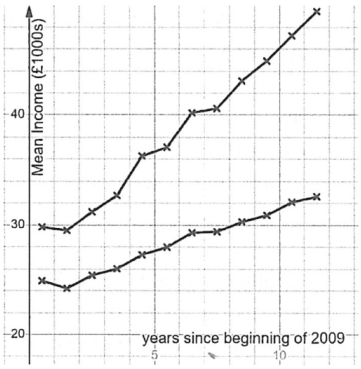
d) Compare these to the statistics calculated above.

1.62 — Question 7. The mean income of taxpayers and median income of taxpayers in 2018/19 for Lewisham and for Kensington and Chelsea are shown in the table below:

	Mean (£)	Median (£)
Kensington and Chelsea	173 000	39 300
Lewisham	38 000	27 900

Compare the distributions of the income of taxpayers for both areas.

1.62 — Question 8. The mean income I in £1000s has been recorded against time t for 12 consecutive years for both Hackney and the North West of England.



a) Identify which of the graphs is Hackney.

The two lines may be modelled by:

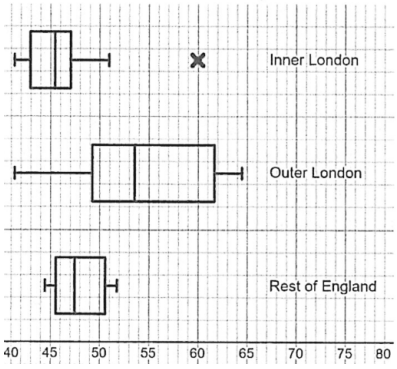
$$I = 1.88t + 26.3 \text{ and } I = 0.781t + 23.3$$

b) Predict the mean income for 2021/2022 according to these models.

c) Explain what the values 1.88 and 26.3, and 0.781 and 23.3, within the models above mean in context.

d) Compare the change in mean income of the two areas.

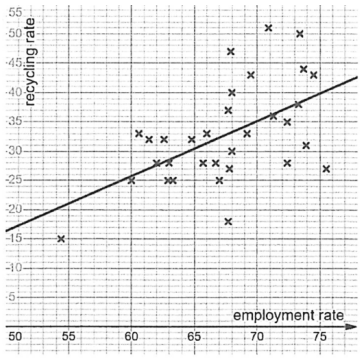
1.62 — Question 9. The percentage of all pupils at the end of KS4 achieving 5 or more A*- C grades including English and Maths in the academic year 2007/8 for Inner London, Outer London and the rest of England are shown in the boxplots below.



a) Using your knowledge of the Large Data Set, identify the name of the outlier in Inner London.
b) Compare the 3 data sets.

1.62 — Question 10. The scatter graph below shows the household recycling rate (%) against the employment rate for 16-64 year olds for 20 of the London boroughs in 2011. The product moment correlation coefficient is $r = 0.7227$ and the least squares regression line is given by

$$y = 0.979x - 33.0$$



a) What does the value of r mean for the association between the household recycling rate and the employment rate?

One of the boroughs not included in the graph has an employment rate of 73.4%.

b) Find the recycling rate of this borough according to the regression line.

Lewisham was not included in the sample. It had an employment rate of 67.7% and a recycling rate of 18% in 2011.

c) If this data point were to be included, explain how the regression line and the value of r would change.

1.62 — Question 11. Prerequisite knowledge for Q11: Normal Distribution The percentage of all pupils achieving 5 or more A*- C grades across the whole of England in 2011/2 may be considered normally distributed, where $X \sim \mathrm{N}(59.4, 4)$. Find

a) the probability that the percentage is higher than 58%.
b) the probability that the percentage is less than 55%.
c) $\mathrm{P}(\mu - 2\sigma < X < \mu + 2\sigma)$

1.63 — Question 1. State the following facts about the Large Data Set:

a) the 10 different categories of method of travel to work (excluding 'other').
b) the extra category to ensure that all people are included in the travel data collection.
c) the 2 different years from which the data was collected.

1.63 — Question 2. What happened to the authorities between 2001 and 2011 which has an impact when comparing the two years with this data?

1.63 — Question 3. Kyra states that if she wants to find all the people who come to work by park and ride the only category she needs to consider is "bus, minibus or coach". Explain why Kyra is wrong.

1.63 — Question 4. In 2011, Abdullah was unemployed the week before the census, but started a new job close to his house the day he filled it in. Which category should he have identified as?

1.63 — Question 5. Toby is considering the 2011 age data by authority. He notices that Exeter has 14155 people aged 20-24 whilst Ashford has 6245 aged 20-24 despite a similar total number of residents. Explain a reason why this might be the case.

1.63 — Question 6. Jakub states that in 2001, the number of people using the Underground, metro, light rail and tram in the South East and the South West can be summarised using the following statistics:

$$\sum x = 1915, \quad \sum x^2 = 196\,261, \text{ and } n = 36$$

a) Calculate the mean and the standard deviation, where $s_x = \sqrt{\dfrac{\sum x^2 - \frac{(\sum x)^2}{n}}{n-1}}$, giving your answer to 1 decimal place.

Jakub's teacher explains he is wrong as he has missed a data point.

b) Calculate the new mean and standard deviation which includes this data point.

1.63 — Question 7. State how a baby which is less than 1 year is categorised in the Large Data Set.

1.63 — Question 8. For each of the following, given a sample of number of cyclists per authority is being taken from the Large Data Set, write down the name of the sampling method used.

a) The order of the authorities is randomised, then every 100th is noted.
b) The order of the authorities is randomised, then the first 5 from each region is noted to obtain 50 authorities.
c) The data set is sorted by authorities and the proportion of each authority is calculated. Using a random number generator and the position in the spreadsheet, each authority is selected to the same proportion as the full data set.
d) The order of the authorities is randomised, then the first 20 authorities are chosen.

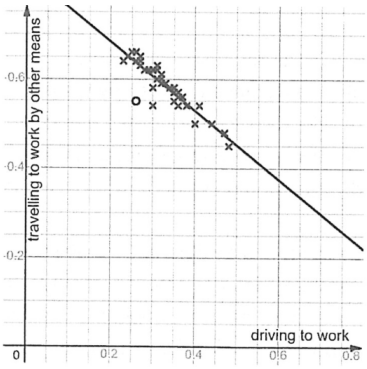
1.63 — Question 9. The table below summarises some age data about London.

	2001	2011
Age 0 to 4	478187	591495
Age 5 to 14	887190	939674
Age 15 to 24	947810	1101631
Age 25 to 44	2533089	2903920
Age 45 to 64	1434225	1732472
Age 65 to 74	468067	473058
Age 75 to 84	310553	308661
Age 85 to 94	112970	123030
	7172091	8173941

a) Is there a significant increase in the proportion of 25–44-year-olds living in London?
b) By comparing the number of 15-24-year-olds living in London in 2011, to the 5-14-year-olds living in London in 2001, comment on the change in this demographic.
c) Explain why the final category includes data outside of the category range.

1.63 — Question 10. The scatter graph below shows the proportion of working people "travelling to work by any means which is not driving" against people "driving a car" in the North West in 2001.
The product moment correlation coefficient, r, is -0.899 to 3 significant figures.

a) One of the points is identified with a circle rather than a cross.
By considering this point, explain why all the points do not lie exactly in a line with $r = 1$.



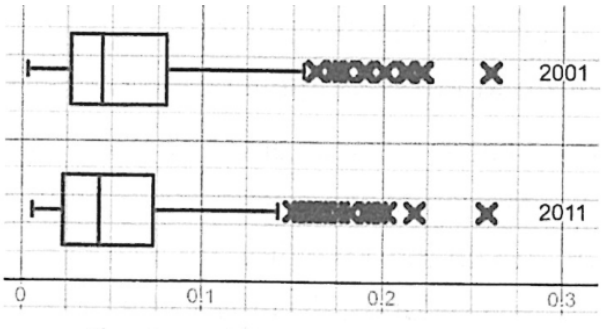
The least squares regression line is given by

$$y = -0.7718x + 0.8374$$

where y is the proportion of people not travelling to work by driving and x is the proportion driving to work.

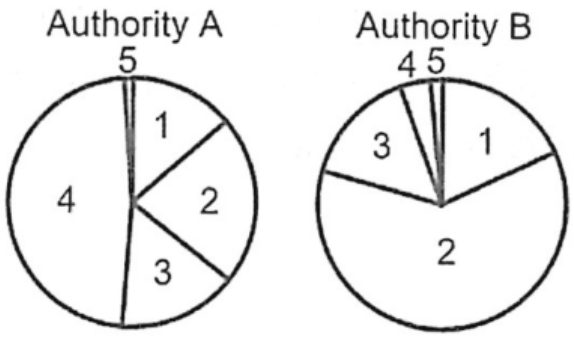
b) By using the regression line, calculate how many of the people in the North West would typically drive to work if they were in a local authority where 50% of people drive to work.

1.63 — Question 11. The proportion of people travelling by bus, minibus or coach in 2001 and 2011 is shown in the boxplots below.



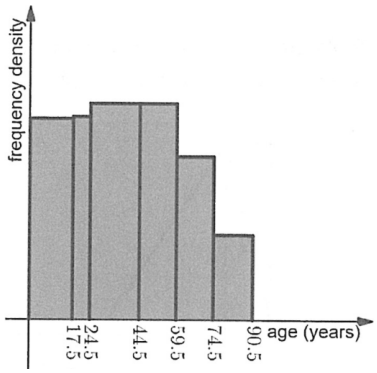
Compare the two data sets.

1.63 — Question 12. Consider the two pie charts below showing data from 2001. One is in London and the other is in a rural location.



The categories are 'any form of public transport', 'working from home', by 'bicycle or on foot', 'car as a passenger or a driver' and 'other' but not in that order.
Identify which is which and make suggestions on which categories 1-5 will be, given they represent the same type of data in both pie charts.

1.63 — Question 13. Prerequisite knowledge for Q13: Normal Distribution Alex creates a histogram to show the distribution of the age of people in the South East in 2011.
There are 8 634 750 people living in this region in 2011.



The heights of the bars are $\{213, 215, 229, 229, 173, 90\}$

a) Estimate the mean and the standard deviation of the data.
b) Estimate the number of people aged 68 or above. Explain why this is an estimation.
c) Would it be reasonable to assume that this data is normally distributed? Fully justify your answer.
d) Explain how the histogram created by Alex could be improved to better represent the data.